

STEFAN OLIVIER WAKATA

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ABOUT ME

Recent Mathematics & Computer Science graduate (University of Ottawa) with hands-on experience building data-driven applications, from machine learning models to full-stack web platforms. Passionate about applying AI and data science to solve real-world problems with measurable impact, including environmental monitoring projects presented to government stakeholders. Fluent in English and French, with a working knowledge of Spanish.

PROFESSIONAL EXPERIENCE

Teaching Assistant — CSI 4107: Information Retrieval & the Internet | University of Ottawa (Contract, Part-time)
Jan 2026 – Apr 2026

- Support undergraduate students by responding to questions on course material, assignments, and projects via email and online channels
- Assist with grading assignments and exams, providing constructive feedback aligned with course rubrics

Research Intern | Nova Scotia Community College, Applied Research
May 2023 – Aug 2023

- Collaborated with a research team to develop a UV fluorescence-based system for real-time detection of oil contamination in aquatic environments
- Assisted with optimizing the UV illumination process and calibrating sensors to improve detection sensitivity, contributing to an innovative solution for environmental monitoring

Sales Associate | H&M
May 2024 – Present

- Provide customer service in a fast-paced retail environment, helping customers find products and resolve issues
- Support inventory management and visual merchandising to help meet store sales targets

PERSONAL PROJECTS

Groundwater Well Monitoring System — Python, Streamlit, SQLite, Plotly, Anthropic API | [Live App](#)

- Collaborated with researchers on the NSCC-CCSIF project to build a real-time web application monitoring 14 groundwater wells across Nova Scotia, visualizing sensor data (water level, temperature, battery) and lab water quality results
- Integrated an AI assistant powered by the Anthropic API to answer natural language questions in English and French, querying a live database and analyzing survey data trends
- Implemented automated flagging of water quality exceedances against Canadian Drinking Water Quality Guidelines (e.g., E. coli, arsenic, pH, manganese); project presented to government stakeholders, including elected officials

Maternal Health Data Analysis & Risk Prediction — Python, Pandas, Matplotlib, Scikit-learn | [GitHub](#)

- Developed and evaluated classification models (Logistic Regression, Random Forest) to predict maternal health risk (low, medium, high) from clinical parameters across 1,000+ patient records
- Achieved 86% accuracy and 0.87 F1-score with a Random Forest model, and used SHAP analysis to identify key risk factors for interpretability

Time Chat Translator — React 18 · TypeScript · Vite · Express.js · Supabase · Docker · Azure · GitHub Actions | [GitHub](#) | [Live App](#)

- Architected and deployed a full-stack real-time multilingual chat platform on Azure (Static Web Apps + App Service) with WebSocket-based live messaging via Supabase Realtime and automatic translation into 10+ languages proxied through a custom Express.js backend API
- Secured the backend with Helmet.js, Zod input validation, Winston logging, and rate limiting; containerized both services with Docker multi-stage builds and automated CI/CD via GitHub Actions (ESLint → Vitest → build → deploy)
- Implemented Row-Level Security on all PostgreSQL tables, 35+ topic channels with live sync, and a custom dark glass morphism UI with an interactive canvas particle system built from scratch

EDUCATION

B.Sc. with Honours in Computer Science and Mathematics
September 2021 – April 2026

University of Ottawa

CERTIFICATIONS

Exploratory Data Analysis for Machine Learning — IBM / Coursera
Issued Sep 2025

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, C, HTML/CSS, Go, SQL

Databases: PostgreSQL, SQLite, Supabase

Frameworks & Tools: React 18, Express.js, Vite, FastAPI, Streamlit, scikit-learn, TensorFlow, Pandas, NumPy, Plotly, Tailwind CSS, Vitest

Development Environments: Docker, GitHub Actions (CI/CD), Azure (Static Web Apps, App Service), Android Studio, Oracle VM, Git/GitHub

Specialized Technologies: Artificial Intelligence, Machine Learning, Data Analysis, Anthropic API